

Topic 1.8 - Rational Functions and Zeros

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

A numerator value of zero creates a rational-function zero only when the denominator is nonzero in the original function. Simplification may reveal true zeros but can also hide excluded inputs. Multiplicity controls whether the graph crosses or touches the axis.

Learning targets

- Determine rational-function zeros while preserving original domain restrictions.
- Use multiplicity and sign charts to describe behavior near zeros.
- Construct rational functions from specified zeros, holes, asymptotes, and scale conditions.

Language and structure

Term	Meaning in this topic
rational zero	An allowed input where the numerator is zero and denominator is nonzero.
excluded value	An input that makes the original denominator zero.
canceled factor	A common factor whose input remains excluded after simplification.
zero multiplicity	The exponent of an uncanceled numerator factor.
sign chart	An interval analysis of whether function values are positive or negative.

Launch - reason before calculating

A student says $x = 3$ is a zero of $R(x) = \frac{(x-3)(x+1)}{(x-3)(x-5)}$ because it makes the numerator zero. Evaluate the claim and identify the actual zero.

Concept 1: Zeros after simplification, restrictions retained

Core idea

Factor numerator and denominator completely. Cancel common factors to reveal the simplified rule, but list original exclusions first. A zero must come from an uncanceled numerator factor and must remain in the domain.

Worked example - annotate the reasoning

Problem. Find all zeros and exclusions of $F(x) = \frac{(x-2)(x+4)}{(x+4)(x-7)}$.

Model reasoning. Exclusions are $x = -4, 7$. Simplified rule $(x-2)/(x-7)$ has zero $x = 2$. The canceled value -4 is a hole.

Check your understanding

Independent

No calculator

Find zeros and exclusions of $G(x) = \frac{x^2-9}{x^2-x-6}$.

Concept 2: Multiplicity and sign at zeros

Core idea

At an odd-multiplicity zero, the rational function changes sign if no other factor changes there; at an even-multiplicity zero it keeps sign and touches the axis. A sign chart should include zeros and vertical asymptotes as interval boundaries, while holes do not necessarily change sign.

Worked example - annotate the reasoning

Problem. Analyze zeros of $S(x) = \frac{(x+2)^2(x-1)}{(x-4)^3}$.

Model reasoning. Zeros are -2 multiplicity 2 (touch/no sign change) and 1 multiplicity 1 (cross/sign change). $x = 4$ is a vertical asymptote, not a zero.

Check your understanding

Independent

No calculator

Describe zero behavior of $T(x) = \frac{(x-5)^3}{(x^2+1)(x+2)^2}$.

Concept 3: Constructing rational functions

Core idea

Use numerator factors for zeros, denominator factors for vertical asymptotes, common factors for holes, and positive quadratics when degree is needed without adding real features. A given point determines the scale factor.

Worked example - annotate the reasoning

Problem. Construct a function with zero 2 simple, zero -3 double, vertical asymptotes 1, -4 , and value $R(0) = 9$.

Model reasoning. Let $R = k(x - 2)(x + 3)^2 / [(x - 1)(x + 4)]$. At 0, the factor ratio is $18/4 = 4.5$, so $k = 2$.

Check your understanding

Independent

No calculator

Construct a rational function with zero 1 double, vertical asymptote -2 , and no other real zeros or asymptotes.

Representation laboratory

Connect at least two representations

A graph crosses the x -axis at -1 , touches at 3, has a hole at 2, and has a vertical asymptote at -4 . Write one factored rational function consistent with these features and state every original exclusion.

Challenge laboratory

Challenge - justify every claim

Construct a rational function with horizontal asymptote $y = 2$, zero -2 of multiplicity 2, zero 3 simple, hole at 1, and vertical asymptote at -4 , with no other real features.

Error clinic and calculator strategy

Common errors to diagnose

- Calling every numerator root a zero without checking the original denominator.
- Forgetting that canceled inputs remain excluded.
- Using denominator roots as x -intercepts.
- Ignoring zero multiplicity in sign and graph behavior.

Technology decision

A graphing utility may display a canceled hole poorly or fill it visually. Factor algebraically, inspect a table near the excluded input, and graph the original and simplified rules together. Zeros should be verified by substitution into the original domain.

AP-style synthesis**Multi-part reasoning task**

Let $F(x) = \frac{(x-4)^2(x+1)}{(x-2)(x+1)(x^2+1)}$.

A. State the domain exclusions and simplified rule.

B. Identify all zeros with multiplicity and graph behavior.

C. Determine the sign of F on each interval formed by its real zero and vertical asymptote.

Exit ticket**Before you leave**

1. What two conditions must an input satisfy to be a rational zero?

2. What does even zero multiplicity do to sign?

3. Why must restrictions be recorded before cancellation?
